TASK SPECIFICATION: P8-01 PHASE P8 — SECURITY HARDENING, COMPLIANCE & QUALITY ASSURANCE

Comprehensive Security Audit (OWASP ASVS, Dependency Scan & Headers)

Git Branch: `test/P8-01-security-review` · Target Pull Request: test/P8-01-security-review

Core Objective & Technical Scope:
Security vulnerability assessment following OWASP ASVS guidelines, OWASP dependency vulnerability scanning, and HTTP security header hardening.

Priority Level	High	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`test/P8-01-security-review`
Upstream Dependencies	P7-06		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Execute OWASP ASVS Level 2 verification audit.
2. Run automated dependency vulnerability scans across Maven and npm.
3. Harden Content-Security-Policy, HSTS, X-Content-Type-Options, and CORS headers.
4. Author penetration test suite covering injection and broken authorization.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `security configuration`
- `docs/security-review.md`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Zero High or Critical vulnerabilities in application dependencies.
- Security headers present and verified on all HTTP responses.

VERIFICATION & TEST COVERAGE STRATEGY

- DependencyCheck scan
- SecurityHeaderTest
- PenetrationTestSuite